

Assignment 9.

1.

WX \ YZ	00	01	11	10
00	1	0	1	0
01	X	1	X	1
11	1	0	1	0
10	X	1	X	1

令 $X = 1$

质蕴含项 $\{w'x', y'z, wx, yz'\}$

实质蕴含项 $\{w'x', y'z, wx, yz'\}$

$$f(w, x, y, z) = w'x' + y'z + wx + yz'$$

		AB			
		00	01	11	10
CD	00	1	1	0	0
	01	1	X	0	0
	11	0	0	X	1
	10	0	0	1	1

令 $X = 1$

质蕴含项: $\{A'C', AC\}$

实质蕴含项 $\{A'C', AC\}$

$$f(A, B, C, D) = AC + A'C'$$

2. (1)

列出所有的最小项

最小项	ABCD
1	0001 ✓
2	0010 ✓
4	0100 ✓
8	1000 ✓
组 1 (一个1)	
3	0011 ✓
5	0101 ✓
6	0110 ✓
9	1001 ✓
10	1010 ✓
12	1100 ✓
组 2 (两个1)	
7	0111 ✓
11	1011 ✓
组 3 (三个1)	

生成质蕴涵项

最小项	ABCD	最小项	ABCD
1, 3	00_1 ✓	3, 7	0_11 ✓
1, 5	0_01 ✓	3, 11	_011 ✓
1, 9	_001 ✓	5, 7	01_1 ✓
2, 3	001_ ✓	6, 7	011_ ✓
2, 6	0_10 ✓	9, 11	10_1 ✓
2, 10	_010 ✓	10, 11	101_ ✓
4, 5	010_ ✓		
4, 6	01_0 ✓		
4, 12	_100 PI_7		
8, 9	100_ ✓		
8, 10	10_0 ✓		
8, 12	1_00 PI_8		

最小项	ABCD
1, 3, 5, 7	0 _ _ P _I ₁
1, 3, 9, 11	_ 0 _ P _I ₂
2, 3, 6, 7	0 _ _ P _I ₃
2, 3, 10, 11	_ 0 _ P _I ₄
4, 5, 6, 7	0 _ _ P _I ₅
8, 9, 10, 11	1 0 _ _ P _I ₆

构造质蕴涵项图.

			X				X				X	
	1	2	3	4	5	6	7	8	9	10	11	12
PI ₁	X		X		X		X					
PI ₂	X		X						X		X	
PI ₃		X	X			X	X					
PI ₄		X	X							X	X	
PI ₅				X	X	X	X					
PI ₆								X	X	X	X	
PI ₇				X								X
PI ₈								X				X

简化后的蕴涵项图

	✓ 1	✓ 2	✓ 4	✓ 5	✓ 6	✓ 8	✓ 9	✓ 10	✓ 12
PI ₁	X			X					
PI₂	X						X		
PI ₃		X			X				
PI₄		X						X	
PI₅			X	X	X				
PI₆						X	X	X	
PI ₇			X						X
PI₈						X			X

$$f(A, B, C, D) = B'D + B'C + A'B + AC'D'$$

答案不唯一

循环图，先选一个包含最小项数目最

多的质蕴涵项，划掉对应的行以及

覆盖[√]最小项所在的列，再重复简化。
的

过程...

(2)

最小项	ABCD	最小项	ABCD	最小项	ABCD
0 ✓	0000	0, 1 ✓	000_	0, 1, 2, 3	00__ PI_1
1 ✓	0001	0, 2 ✓	00_0	1, 3, 9, 11	_0_ PI_2
2 ✓	0010	0, 4 PI_7	0_00	2, 3, 10, 11	_0 PI_3
4 ✓	0100	1, 3 ✓	00_1	9, 11, 13, 15	1__ PI_4
3 ✓	0011	1, 9 ✓	_001	10, 11, 14, 15	1_ PI_5
9 ✓	1001	2, 3 ✓	001_	12, 13, 14, 15	11- PI_6
10 ✓	1010	2, 10 ✓	_010		
12 ✓	1100	4, 12 PI_8	_100		
11 ✓	1011	3, 11 ✓	_011		
13 ✓	1101	9, 11 ✓	10_1		
14 ✓	1110	9, 13 ✓	1_01		
15 ✓	1111	10, 11 ✓	101_		
		10, 14 ✓	1_10		
		12, 13 ✓	110_		
		12, 14 ✓	11_0		
		11, 15 ✓	1_11		
		13, 15 ✓	1_11		
		14, 15 ✓	111_		

	1	2	3	4	9	10	11	12
PZ_1	X	X	X					
PZ_2 ✓	X		X		X		X	
PZ_3 ✓		X	X			X	X	
PZ_4					X		X	
PZ_5						X	X	
PZ_6								X
PZ_7				X				
PZ_8 ✓				X				X

$$f(A, B, C, D) = B'D + B'C + BC'D'$$